DGMS JSO GEOLOGY 2019 • SEDIMENTARY GEOCHEMISTRY & MARINE GEOLOGY

Question 96: Halmyrolysis (Submarine Weathering & Ion Exchange)

Geochemical Mechanism, River-Ocean Interface Reactions, and Sedimentary Authigenesis

Q.96: Halmyrolysis refers to a process where

(a) Addition of La takes place in magma during crustal contamination

(b) Removal of Mg from magma takes place during crystal fractionation

✓ **(c) Removal of some ions of river water takes place when the river enters the sea**

(d) Removal of Eu takes place during magmatic crystallization

Correct Option: (c) Removal of some ions of river water takes place when the river enters the sea

1. Geological Definition of Halmyrolysis

The term **halmyrolysis** (derived from Greek *halme* = sea salt / brine, and *lysis* = dissolution / decomposition) was coined by Hummel (1922) to denote **submarine weathering**. It encompasses the collective geochemical and mineralogical alterations that affect sediments, clay minerals, and volcanic materials on the seafloor as a result of their direct interaction with seawater prior to deep burial and diagenesis.

A critical aspect of halmyrolysis occurs at the estuary-marine interface. When terrestrial freshwater discharged by rivers enters the high-ionic-strength marine environment:

- **Flocculation & Precipitation:** Colloidal particles of iron, aluminium, and organic complexes destabilize due to the compression of the electrical double layer, causing them to aggregate and settle out of the water column.
- **Cation Exchange & Ion Scavenging:** Detrital clay minerals carried by river water (which are rich in adsorbed Ca^{2+}) undergo rapid ion exchange with seawater cations. The clay surfaces selectively take up seawater potassium (K^{+}), magnesium (Mg^{2+}), and sodium (Na^{+}), while releasing calcium (Ca^{2+}). This effectively removes critical dissolved ions from solution.
- **Authigenic Mineral Formation:** Submarine alteration of basaltic glasses and volcanic ashes produces authigenic smectite, phillipsite (zeolite), and green pelletal glauconite in shallow-to-intermediate marine shelf settings.

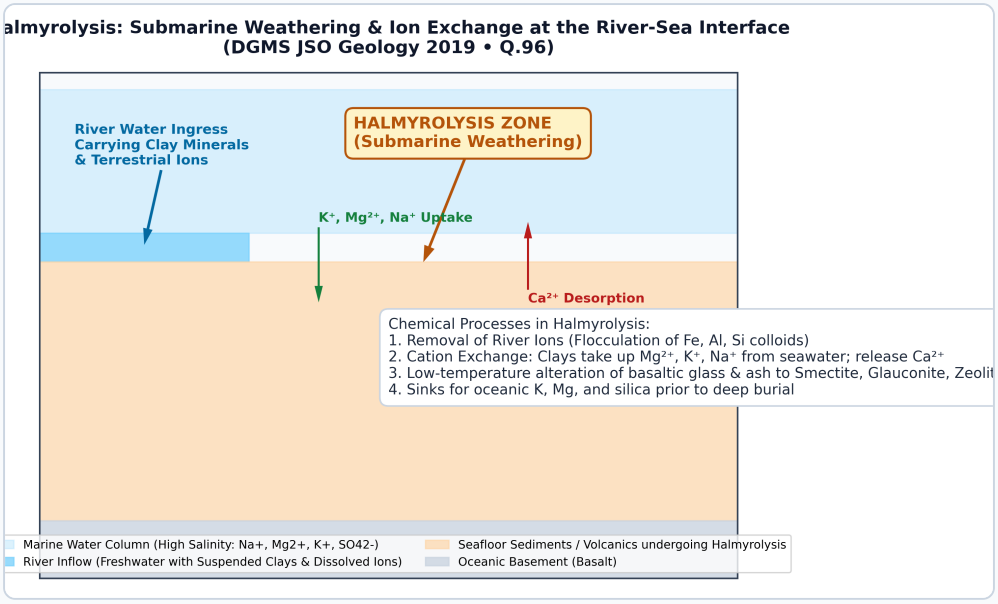


Figure 96.1: Conceptual geochemical cross-section of halmyrolysis at the river-sea boundary showing flocculation, clay cation exchange (uptake of K^{+}/Mg^{2+}/Na^{+}, desorption of Ca^{2+}), and submarine weathering of seafloor sediments. Based on Boggs, Principles of Sedimentology and Stratigraphy.

2. Systematic Option Evaluation

Option	Stated Process	Geochemical Domain	Evaluation & Physical Mechanism
(a)	Addition of La takes place in magma during crustal contamination	Igneous Petrology / Geochemistry	**Incorrect.** This is crustal assimilation (AFC process), where incompatible light rare earth elements (LREE like Lanthanum) are enriched in ascending magmas by assimilating continental crust.
(b)	Removal of Mg from magma takes place during crystal fractionation	Igneous Petrology	**Incorrect.** This describes fractional crystallization of magnesium-rich ferromagnesian phases (olivine, clinopyroxene) during Bowen's reaction series.
(c)	**Removal of some ions of river water takes place when the river enters the sea**	**Sedimentary Geochemistry / Marine Geology**	**Correct.** By definition, halmyrolysis involves the chemical interaction between river-borne sediments/ions and seawater, leading to cation exchange, colloidal flocculation, and the removal of dissolved ions from water.
(d)	Removal of Eu takes place during magmatic crystallization	Igneous Trace Element Geochemistry	**Incorrect.** This describes the formation of a negative europium anomaly ($\delta Eu < 1$) due to preferential

Option	Stated Process	Geochemical Domain	Evaluation & Physical Mechanism
			partitioning of divalent Eu^{2+} into fractionating plagioclase feldspar.

> **Key Geochemical Takeaway**
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> While subaerial weathering is driven by meteoric water, carbonic acid, and atmospheric oxygen, **halmyrolysis** is submarine chemical weathering driven by saline ocean water. It acts as a primary geochemical sink for alkali and alkaline earth cations (especially K^+ and Mg^{2+}) and regulates the chemical composition of the oceans.

3. Standard Textbook References

- **Sam Boggs Jr.**, *Principles of Sedimentology and Stratigraphy* (5th Edition, Pearson, 2012), Chapter 5: "Chemical/Biochemical Sedimentary Rocks", pp. 142–145.
- **F. J. Pettijohn**, *Sedimentary Rocks* (3rd Edition, Harper & Row), Chapter 11: "Diagenesis and Weathering", Section: Halmyrolysis, pp. 418–422.
- **Robert A. Berner**, *Principles of Chemical Sedimentology* (McGraw-Hill, 1971), Chapter 8: "Ion-Exchange and Early Marine Diagenesis", pp. 165–180.